

Sandra Tang

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EDUCATION

Cornell University, College of Engineering

Ithaca, NY

B.S. in Computer Science

Expected May 2028

- **Relevant Coursework:** Systems Programming, Design with Embedded Operating Systems (ECE), Digital Systems Design Using Microcontrollers (ECE), Computer System Organization and Programming, Analysis of Algorithms, Object-Oriented Programming & Data Structures, Computer Vision, Machine Learning, Functional Programming

TECHNICAL EXPERIENCE

Amazon Leo (GNC & Propulsion)

June 2026 – August 2026

Flight Dynamics Software Developer Intern | *Java, TypeScript, AWS CDK*

Redmond, WA

- Built and owned a service that runs astrodynamics engineers' analytics scripts automatically, either on a schedule or in response to satellite events such as an ephemeris update, so work that had been triggered by hand now runs itself; shipped to production for 100+ engineers and architected to scale to 300+ scripts.
- Designed the serverless architecture as code in TypeScript CDK (Lambda, Step Functions, SNS, S3, DynamoDB) and authored the Java handler package implementing trigger, execution, and failure-triage logic that root-causes a broken run and auto-routes the ticket to the script owner or the platform team.
- Took the project through its full lifecycle in a 12-week internship, gathering requirements directly from the engineers who would use the platform rather than working from a spec, then automated a manual, fleet-wide analytics script by triggering it on ephemeris update events across the satellite fleet, eliminating a recurring weekly manual run as the platform's first live use case.

Cornell Unmanned Air Systems

February 2025 – Present

Intelligence Subteam Lead (since August 2026) | *Rust, Python, Embedded Linux, MAVLink*

Ithaca, NY

- Lead a subteam of 15 engineers owning the onboard imaging and telemetry pipeline for an autonomous UAV end to end, aligning 1,300+ GoPro images per 45-minute flight to their GPS and attitude samples within 125 ms using timestamp-based buffering and nearest-sample matching.
- Run the pipeline on a Raspberry Pi with the GoPro and Pixhawk attached over USB, bridged to the ground station over a Ubiquiti Rocket and NanoStation link for SSH access and live telemetry mid-flight.
- Timestamped camera captures and MAVLink telemetry against a single onboard clock instead of reconciling the GoPro's and Pixhawk's, pairing each image with its nearest sample and holding back any image with no sample in tolerance rather than uploading it.
- Validate every change on a staged test ladder (bench, ground rig, test drone, competition aircraft), with a preflight script that checks the camera, Pixhawk, ground server, network, and disk before each flight.
- Served as Intelligence operator on the flightline at SUAS 2026, the international Student Unmanned Aerial Systems competition, solely responsible for the pipeline during live missions; review the subteam's code, onboard new members each semester, and own the data interface with the Autopilot subteam.

Hao Shi Guang Restaurant

June 2025 – Present

Freelance Web Developer | *React, MongoDB*

Boston, MA

- Sole developer of a live restaurant web app (15+ months): menus, MongoDB reservations, Toast ordering.

PROJECTS

Birdsong Synthesizer

Fall 2026

- Built a birdsong synthesizer on an RP2350 in C using direct digital synthesis in a 200 kHz timer interrupt, streamed over SPI to an MCP4822 12-bit DAC configured from its datasheet, with a debounced keypad state machine.

Embedded Linux Touch Controller

Fall 2026

- Built GPIO-button and touchscreen controls on a Raspberry Pi 4 for video playback and a PyGame collision animation, replacing polling with interrupt callbacks and profiling both with Linux perf.

Camel Benchmark

Fall 2025

- Refactored a game module into a pure state machine, decoupling core logic from async I/O so behavior could be tested in isolation, then wrote test suites exercising state transitions, input parsing, and edge cases to reach 100% Bisect coverage.

TECHNICAL SKILLS

Languages: Java, TypeScript, Python, C, C++, Rust, JavaScript, SQL, OCaml, HTML/CSS

Systems & Tools: Embedded Linux, MAVLink, Pixhawk, Bash/Shell, Git, Docker, AWS (CDK, Lambda, Step Functions, SNS, S3, DynamoDB), REST APIs, React.js, Node.js, Flask, Prisma, MongoDB, MySQL, PyTorch, OpenCV, NumPy